



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Final Year B. Tech.

Advance Database Management Systems

ISE-2 Exam (Semester-I) 2025-26

Computer Science and Engineering
PCC-CS401

Date: 14-10-2025 Time: 14.30 to 15.15

Max. Marks: 20 Marks

- Note:
- Attempt all questions.
 - Assume suitable data if necessary.
 - Use of non-programmable calculator is permitted.
 - Write point to point answers in short.

Q. N	Question	Marks	CO	BT
Q.1	Solve any three questions. (3 x 4 = 12 M)	12 M		
a	Draw and Explain Tree-like interconnection architecture.	4 M	CO1	BT2
b	Analyze and Explain layers of distributed query processing with diagram.	4 M	CO2	BT4
c	For the given SQL query write down two equivalent relational algebra queries that are correct transformations of the query. <i>SELECT ENAME FROM EMP, ASG WHERE EMP.ENO = ASG.ENO AND RESP = 'Manager'</i>	4 M	CO3	BT3
d	Explain xml with examples.	4 M	CO5	BT2
Q.2	Solve any two questions. (2 x 4 = 8 M)	8 M		
a	Consider an engineering firm that has offices in Boston, Waterloo, Paris and San Francisco. They run projects at each of these sites and would like to maintain a database of their employees, the projects and other related data. Assuming that the database is relational, we can store this information in two relations: EMP (ENO, ENAME, TITLE)1 and PROJ (PNO, PNAME, BUDGET). We also introduce a third relation to store salary information: SAL (TITLE, AMT) and a fourth relation ASG which indicates which employees have been assigned to which projects for what duration with what responsibility's (ENO, PNO, RESP, DUR). Write down the SQL query to find out the names and employees who worked on a project for more than 24 months.	4 M	CO2	BT2
b	Explain four characteristics of distributed query processor.	4 M	CO2	BT2
c	How does query execution in a Distributed Database Management System (DDBMS) differ from that in a Centralized Database Management System (CDBMS)?	4 M	CO2	BT4

Course Outcomes (CO)

- CO1: Apply concepts of distributed and parallel databases to design efficient database architectures and query processing strategies.
- CO2: Analyze transaction management techniques, including commit protocols, concurrency control, and fault tolerance in distributed and multidata base systems.
- CO3: Develop and evaluate advanced querying and information retrieval methods such as OLAP, data mining, and data warehousing for decision support systems.
- CO4: Demonstrate proficiency in handling advanced data types (spatial, multimedia, temporal, mobile) and their applications in modern database systems.
- CO5: Design and implement applications using emerging technologies such as XML databases, multidimensional indexing, and web-based database interfaces.
- CO6: Critically evaluate performance issues, tuning methods, and administration challenges in advanced DBMS for real-time, large-scale, and heterogeneous environments.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science & Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination II

Class: TY

Course Name: Elective: Advanced
Python Programming

Course Code: PEC-CS303

Date: 14-Oct-25

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- i. Mobile phones and programmable calculators are strictly prohibited.

Q.N.	Attempt All the Questions	Marks	CO	BT
1	Explain the concept of dynamic attributes with suitable example.	6	CO1	BT1
2	Explain the concept of Design Pattern "Singleton" with suitable example.	7	CO4	BT2
3	With a suitable program explain multithreading in python.	7	CO2	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT):

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply advanced object-oriented programming concepts, including meta programming and design patterns.
CO 2	Implement concurrent and parallel programs using threads, processes, and asynchronous techniques.
CO 3	Develop network applications using sockets and advanced networking libraries.
CO 4	Utilize advanced Python libraries for data manipulation, scientific computing, and web development.
CO 5	Write efficient and optimized Python code for performance-critical applications.
CO 6	Design and implement robust and maintainable Python applications.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science & Engineering
In-Semester Examination II
Academic Year: 2025-26 (semester I)

Class: TY

Course: Ethical Hacking

Course Code: OE-CS303

Date: 17/10/2025

Time: 11:00 to 11:45

Max. Marks: 20

Note:

- Assume suitable data whenever necessary and state it clearly.
- Mobile phones and programmable calculators are strictly prohibited.
- Exchange/Sharing of stationery, calculator etc. not allowed.
- Figures to the right indicate full marks.
- Write your answers step-by-step.

Q.N.	Attempt All Questions	Marks	CO	BT
Q.1	Encrypt the plaintext "SECURITY" using the Caesar Cipher technique with a key of 4. Show the encryption process and find the cipher text.	5	CO4	BT3
Q.2	Differentiate between MD5 and SHA	5	CO4	BT2
Q.3	Encrypt the message "CRYPTOGRAPHYTECHNIQUES" using Rail Fence Cipher with 3 rails.	10	CO4	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Understand the fundamental concepts of ethical hacking, computer networking, and TCP/IP protocol stack, including IP addressing, routing, and IPv6.
CO 2	Develop skills in system installation and information gathering techniques using tools like Google search, WHOIS, DNS enumeration, and NMAP for penetration testing.
CO 3	Analyze vulnerabilities through scanning tools like NMAP and Nessus, and learn practical techniques for system hacking, password cracking, and privilege escalation.
CO 4	Gain knowledge of cryptographic techniques including private-key and public-key encryption, cryptographic hash functions, and digital certificates for securing communications.
CO 5	Implement advanced security measures such as malware analysis, ARP spoofing, packet sniffing with Wireshark, and social engineering techniques for network-based attacks.
CO6	Understand hardware security concepts, including side-channel attacks and hardware Trojans, and apply penetration testing methods using tools like Metasploit to perform real-world cyberattack simulations.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

ISE-II Examinations (Semester-I) 2025-26

T.Y. B. Tech.

Computer Science and Engineering

Machine Learning

OE-CS301

Date: 17/10/25

Time: 11.00-11.45

Max. Marks: 20 Marks

Q.N.	Questions (ATTEMPT ALL)	Marks	CO	BT
Q.1	Differentiate between supervised and unsupervised learning.	5	CO2	BT2
Q.2	Define SVM and explain hyperplane, support vectors, and margin with diagram.	5	CO1	BT1
Q.3	Define Ensemble learning? Explain types of Ensemble learning	5	CO3	BT1
Q.4	Given the following 2D data points: P1(2, 10), P2(2, 5), P3(8, 4), P4(5, 8), P5(7, 5), P6(6, 4), Let K = 2, and initial centroids are: C1 = (2, 10), C2 = (5, 8) Perform one iteration of K-Means clustering:	5	CO2	BT3

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO1	Understand the basic concepts, strengths and weaknesses of many popular machine learning approaches.
CO 2	Learn the underlying mathematical relationships within and across Machine Learning algorithms and the paradigms of supervised and un-supervised learning.
CO 3	Apply popular machine learning techniques and computing environment that are suitable for the applications under consideration.
CO 4	Identify and Analyze various ways of selecting suitable model parameters for different machine learning techniques.
CO 5	Design and implement various machine learning algorithms in a range of real-world applications.
CO 6	Evaluate the performance of machine learning models using appropriate metrics and techniques to ensure reliability and accuracy in diverse problem-solving scenarios.

	SGGS Institute of Engineering and Technology, Vishnupuri, Nanded	
	ISE-II Examinations (Semester-I) 2025-26	
	T.Y. B. Tech.	Computer Science and Engineering
	Database management system	MDM-CS301
Date: 16/10/25	Time: 11.00-11.45	Max. Marks: 20 Marks

Q.N	Questions (ATTEMPT ALL)	Marks	CO	BT
Q.1	What is normalization? Explain 1NF, 2NF & 3NF with example.	5	CO4	BT1
Q.2	Explain different types of integrity constraints.	5	CO3	BT2
Q.3	Consider following schema and represent given statements in relation algebra form.(2 marks each) Branch(branch_name, branch_city) Account (acc_no, branch_name, balance) Customer(customer_id, customer_name, customer_city) Depositor(customer_id, acc_no) 1. Find the customer IDs of all customers who live in 'Mumbai' or 'Delhi'. 2. Find the account numbers of accounts with balance between 20,000 and 50,000 3. Find the names of customers along with their account numbers. 4. Find customers who live in 'Pune'. 5. Customer names who live in 'Delhi' but do not have accounts there.	10	CO3	BT3

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Identify and describe the different issues involved in a database system.
CO 2	Construct Entity-Relationship Model for given applications and Relational Model for the same.
CO 3	Design and write best possible or optimal (SQL) query statement for given statement.
CO 4	Apply normalization to database design.
CO 5	Improve efficiency of data retrieval using various storage systems and indexing.
CO 6	Provide overview of transaction management, concurrency control, database recovery and security.



Shri Guru Gobind Singhji Institute of Engg. & Technology, Vishnupuri, Nanded-431606.
Department of Computer Science and Engineering
Academic Year: 2025-26 (semester I)
In-Semester Examination II

Class: Third Year

**Course Name: MDM-
Introduction to Cryptography**

Course Code: MDM-IS-CS301

Date: 16/10/2025

Time: 11.00 to 11.45

Max. Marks: 20

Note:

- i. Solve any two questions.
- ii. Assume suitable data whenever necessary and state it clearly.
- iii. Mobile phones and programmable calculators are strictly prohibited.
- iv. Exchange/Sharing of stationery, calculator etc. not allowed.
- v. Figures to right indicate full marks.
- vi. The symbols carry usual meaning.

Q.N.			Marks	CO	BT
Q.1	a	I. Compare reduced set residues vs complete residue system for $n = 5$.	2	CO1	BT4
		II. Construct addition table for Z_5 .	3		BT3
	b	Explain the role of ShiftRows transformation in AES?	5	CO2	BT2
Q.2	a	I. Find value of $2^{11} \bmod 11$ using Fermat's Little theorem.	2	CO1	BT1
		II. Compute $\phi(8)$ and $\phi(64)$.	3	CO1	BT3
	b	Analyze why Double DES (2DES) is not significantly more secure than DES?	5	CO2	BT4
Q.3	a	What are the main weaknesses of DES?	5	CO2	BT2
	b	Solve: $x \equiv 1 \pmod{4}$, $x \equiv 2 \pmod{5}$, and $x \equiv 3 \pmod{7}$ using Chinese Remainder theorem.	5	CO1	BT3

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

On successful completion of this course, students should be able to:
CO1: Understand the number theory concepts.
CO2: Impart knowledge on symmetric and asymmetric encryption techniques.
CO3: Create Authentication functions as Message Authentication Codes and Hash Functions
CO4: Examine the issues and structure of Authentication Service and Electronic Mail, web and IP security.
CO5: Provide familiarity in system security.
CO6: Understand and apply the principles of cryptographic protocols and network security to ensure secure communication and data protection.